

Monte Carlo and Crank–Nicolson option pricing

Numerical methods and an option-chain application

A comparison of simulation and finite-difference methods for option values and hedge ratios, followed by a market-data application with pricing and risk diagnostics.

Methods	C++; Euler–Maruyama; Crank–Nicolson; Black–Scholes; Python visualisation
Data	Specified CEV-type model; supplied SPY option-chain snapshot, 18 February 2026
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1 Executive summary

This project compares Monte Carlo simulation and Crank–Nicolson finite differences for a European put under a CEV-type diffusion, then applies the numerical methods to a supplied option-chain snapshot.

In the specified diffusion, Monte Carlo gives a put value of 9.0577 with a 95% sampling interval of [8.7989, 9.3165]. Crank–Nicolson gives 9.1295. The corresponding deltas are -0.4009 and -0.4071 , with the finite-difference result inside the Monte Carlo delta interval. An effective-volatility Black–Scholes value of 9.1492 provides an approximate reference for the CEV-type model.

The market application covers 947 SPY contracts across 10 expiries, using a Yahoo Finance snapshot obtained through yfinance. Closed-form Black–Scholes and Crank–Nicolson give similar aggregate prices. Comparisons with quoted mids, together with volatility, Greeks and scenario plots, connect the numerical methods to practical pricing and risk analysis.

The two parts bring together model formulation, numerical implementation and interpretation of price sensitivities. The report compares sampling behaviour, grid refinement and option-chain outputs within a common pricing workflow.

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2 Data and scope

2.1 Project origin and scope

This work adapts a project completed during the MSc in Financial Mathematics at King’s College London. The finite-difference and Monte Carlo methods build on John Armstrong’s Computational and Numerical Methods in Mathematical Finance lectures [1].

Part A compares numerical methods under a specified CEV-type model. Part B applies European Black–Scholes, Monte Carlo and PDE calculations to an option-chain snapshot. These are distinct modelling settings: the effective-volatility reference in Part A is approximate, whereas Black–Scholes is an exact closed-form reference for Part B’s assumed European GBM model.

2.2 Specified diffusion: Part A

It is given that a stock price S_t evolves according to

$$dS_t = \mu S_t dt + \sigma(t) S_t^\gamma dW_t.$$

However, for pricing we work under the risk-neutral measure $\mathbb{Q}$, where the drift becomes the risk-free rate r (under standard no-arbitrage assumptions). Hence the dynamics we use is:

$$dS_t = r S_t dt + \sigma(t) S_t^\gamma dW_t,$$

where W_t denotes a $\mathbb{Q}$ -Brownian motion.

The parameters are $\gamma = 0.95$, $\sigma(t) = 0.18 + 0.07 \cdot \frac{t}{T}$, $K = 140$, $r = 0.02$, $S_0 = 140$, and $T = 1.40$.

The goal in Part A is to compute the European put price $V(S, t)$ and delta Δ under the risk-neutral dynamics, then compare two numerical approaches:

- Monte Carlo simulation using Euler–Maruyama,
- Crank–Nicolson finite differences applied to the pricing PDE.

To benchmark the outputs, we also compute an *estimated* Black–Scholes value using an effective volatility σ_{eff} , since the true model here is not log-normal.

Part B uses market strikes, expiries, quotes and implied volatilities as inputs to a log-normal European pricing model with continuous dividend yield. The purpose is to compare numerical implementations and inspect model–market differences.

2.3 Market snapshot: Part B

The market snapshot contains SPY options sourced from Yahoo Finance through yfinance. It is timestamped 18 February 2026 at 02:59:31 UTC, with spot 682.85 and dividend yield 1.050%. The cleaned chain contains 947 contracts: 280 calls and 667 puts.

Prices are expressed in the supplied USD quote units. The chain includes strikes, expiries, market mids and implied volatilities, allowing the three numerical approaches to be compared using common contract inputs.

3 Methodology

3.1 Pricing PDE and hedging argument

Consider the option value $V(S, t)$ and form a self-financing hedged portfolio by shorting one option and holding Δ shares:

$$\Pi(t) = -V(S_t, t) + \Delta_t S_t.$$

The portfolio change is

$$d\Pi = -dV + \Delta dS.$$

Using Itô's lemma on $V(S_t, t)$ under the risk-neutral dynamics:

$$dV = V_t dt + V_S dS + \frac{1}{2} V_{SS} (dS)^2.$$

Since $dS = rS dt + \sigma(t)S^\gamma dW$, we have $(dS)^2 = \sigma^2(t)S^{2\gamma} dt$ (and the mixed terms vanish). Substituting gives:

$$dV = \left(V_t + rSV_S + \frac{1}{2} \sigma^2(t)S^{2\gamma} V_{SS} \right) dt + \sigma(t)S^\gamma V_S dW.$$

Now plug this into the portfolio dynamics. If we choose the hedge ratio $\Delta = V_S$, the dW terms cancel (the local diffusion risk is hedged), and the portfolio becomes locally risk-free:

$$d\Pi = - \left(V_t + \frac{1}{2} \sigma^2(t)S^{2\gamma} V_{SS} \right) dt.$$

Under no-arbitrage in the risk-neutral world, a risk-free portfolio must earn the risk-free rate:

$$d\Pi = r\Pi dt = r(-V + SV_S) dt.$$

Equating these and rearranging yields the pricing PDE:

$$V_t + rSV_S + \frac{1}{2} \sigma^2(t)S^{2\gamma} V_{SS} - rV = 0,$$

with terminal condition for a European put:

$$V(S, T) = (K - S)^+.$$

This PDE is what we discretise in the Crank–Nicolson method, and it also links to Monte Carlo via Feynman–Kac.

For the market-data application (Part B), the same hedging logic produces the standard Black–Scholes PDE under GBM, with the minor modification that a continuous dividend yield q enters the drift and boundary behaviour. In that setting we use Black–Scholes as a baseline model, Monte Carlo under GBM as a probabilistic cross-check, and Crank–Nicolson as the PDE-based solver.

3.2 Monte Carlo simulation

By Feynman–Kac, the put price can be written as a discounted expectation:

$$V(S, t) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} (K - S_T)^+ \mid S_t = S \right].$$

Monte Carlo approximates this expectation by simulating many paths of S_t and averaging the discounted payoff.

The law of large numbers justifies averaging independent discounted payoffs. With finite payoff variance, the central limit theorem gives sampling error of order $\mathcal{O}(N^{-1/2})$. At a fixed Euler time step, this convergence is to the discretised simulation expectation; increasing the path count alone does not remove time-discretisation bias.

To simulate paths under the risk-neutral SDE, we discretise using Euler–Maruyama with $\Delta t = T/n$:

$$S_{i+1} = S_i + r S_i \Delta t + \sigma(t_i) S_i^\gamma \sqrt{\Delta t} \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1).$$

In the computations we used $n = 100$ time steps and $N = 10,000$ simulated paths.

The Euler step evaluates the fractional diffusion term using $\max(S, 0)^\gamma$. The drift and simulated state are updated using the Euler–Maruyama scheme above.

3.2.1 Price and sampling interval

The Monte Carlo estimator is:

$$\hat{V} = e^{-rT} \frac{1}{N} \sum_{j=1}^N (K - S_T^{(j)})^+.$$

A 95% confidence interval is:

$$\hat{V} \pm z_{0.975} \frac{\hat{s}}{\sqrt{N}},$$

where $\hat{s}$ is the sample standard deviation of discounted payoffs.

This interval describes sampling uncertainty under the simulated model. It excludes discretisation bias, parameter uncertainty and model misspecification.

3.2.2 Delta estimation

Delta is defined as $\Delta = \frac{\partial V}{\partial S}$. We approximate it using a central difference:

$$\Delta \approx \frac{V(S_0 + \epsilon) - V(S_0 - \epsilon)}{2\epsilon},$$

with $\epsilon = 0.01 S_0$. To reduce variance, we used common random numbers when estimating $V(S_0 + \epsilon)$ and $V(S_0 - \epsilon)$.

3.3 Crank–Nicolson finite differences

The Crank–Nicolson method blends explicit and implicit schemes and takes their average. In practice, this gives a second-order accurate time discretisation (in the smooth regime) and is typically much more stable than a purely explicit method.

Like the Monte Carlo method, discretisation is required. However, here we discretise the pricing PDE directly on a grid. We choose a spatial grid

$$S_i = i \delta S, \quad i = 0, 1, \dots, M,$$

and a time grid

$$t_j = j \delta t, \quad j = 0, 1, \dots, P, \quad \delta t = \frac{T}{P}.$$

We write $V_i(t) \approx V(S_i, t)$, and at each fixed t we apply central differences in S :

$$V_S(S_i, t) \approx \frac{V_{i+1}(t) - V_{i-1}(t)}{2 \delta S}, \quad V_{SS}(S_i, t) \approx \frac{V_{i+1}(t) - 2V_i(t) + V_{i-1}(t)}{(\delta S)^2}.$$

Substituting these into the PDE and rearranging to isolate the time derivative gives a semi-discrete ODE system for each interior node $i = 1, \dots, M - 1$:

$$\frac{dV_i}{dt} + \frac{1}{2} \sigma^2(t) (S_i)^{2\gamma} \frac{V_{i+1} - 2V_i + V_{i-1}}{(\delta S)^2} + r S_i \frac{V_{i+1} - V_{i-1}}{2 \delta S} - r V_i = 0.$$

Equivalently,

$$\frac{dV_i}{dt} = a_i(t) V_{i-1} + b_i(t) V_i + c_i(t) V_{i+1},$$

where the coefficients are:

$$\begin{aligned} a_i(t) &= -\frac{1}{2} \frac{\sigma^2(t) (S_i)^{2\gamma}}{(\delta S)^2} + \frac{r S_i}{2 \delta S} \\ b_i(t) &= r + \frac{\sigma^2(t) (S_i)^{2\gamma}}{(\delta S)^2} \\ c_i(t) &= -\frac{1}{2} \frac{\sigma^2(t) (S_i)^{2\gamma}}{(\delta S)^2} - \frac{r S_i}{2 \delta S} \end{aligned}$$

Let $\mathbf{V}(t)$ denote the vector of interior option values

$$\mathbf{V}(t) = (V_1(t), V_2(t), \dots, V_{M-1}(t))^{\top}.$$

Stacking the semi-discrete equations across all interior nodes yields a linear ODE system of the form:

$$\frac{d\mathbf{V}}{dt} = \Lambda(t) \mathbf{V} + \mathbf{w}(t),$$

where:

$\mathbf{V}(t)$ is the vector of option values at the interior grid asset prices.

$\Lambda(t)$ is the tridiagonal matrix containing the drift and diffusion coefficients $a_i(t), b_i(t), c_i(t)$.

$\mathbf{w}(t)$ is the vector of boundary contributions (only the first and last interior equations depend on the boundary values).

Grid: $(P + 1) \times (M + 1)$ for the full grid, with $(M - 1)$ interior spatial nodes and $(P + 1)$ time levels; $\delta t = T/P$ and $\delta S = S_{\max}/M$.

$$\Lambda(t) = \begin{pmatrix} b_1(t) & c_1(t) & 0 & 0 & \cdots & 0 \\ a_2(t) & b_2(t) & c_2(t) & 0 & \cdots & 0 \\ 0 & a_3(t) & b_3(t) & c_3(t) & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & b_{M-2}(t) & c_{M-2}(t) \\ 0 & 0 & \cdots & 0 & a_{M-1}(t) & b_{M-1}(t) \end{pmatrix}.$$

$$\mathbf{w}(t) = \begin{pmatrix} a_1(t) V_0(t) \\ 0 \\ \vdots \\ 0 \\ c_{M-1}(t) V_M(t) \end{pmatrix},$$

where $V_0(t) = V(0, t)$ and $V_M(t) = V(S_M, t)$ are the boundary values at $S = 0$ and $S = S_{\max}$, respectively. This form makes it clear that the spatial discretisation leads to a tridiagonal linear operator, which is why each Crank–Nicolson time step can be computed efficiently by solving a tridiagonal linear system.

3.3.1 Boundary and terminal conditions

To make the PDE numerically solvable we truncate the spatial domain to $S \in [0, S_{\max}]$, with $S_{\max}$ chosen sufficiently large so that the put is essentially worthless at the top boundary. In the implementation used here we set $S_{\max} = 250$, which is well above $K = 140$ in the reported run.

The boundary and terminal conditions used were:

Top boundary: $V(S_{\max}, t) = 0$.

Bottom boundary: $V(0, t) = Ke^{-r(T-t)}$.

Terminal condition: $V(S, T) = (K - S)^+$.

3.3.2 Time stepping

Crank–Nicolson applies the trapezium rule in time. Writing $\mathbf{V}^j \approx \mathbf{V}(t_j)$, a standard CN discretisation is:

$$\frac{\mathbf{V}^{j+1} - \mathbf{V}^j}{\delta t} = \frac{1}{2} \left(\Lambda^{j+1} \mathbf{V}^{j+1} + \Lambda^j \mathbf{V}^j \right) + \frac{1}{2} (\mathbf{w}^{j+1} + \mathbf{w}^j).$$

Since we solve backwards from the terminal condition at $t = T$, at each step we know $\mathbf{V}^{j+1}$ and solve a tridiagonal linear system to obtain $\mathbf{V}^j$.

The formula above gives the time-centred scheme. In the CEV implementation, the time-dependent coefficients are evaluated at the earlier grid time within each step. The temporal refinement experiment later in the report examines the behaviour of this implementation.

3.3.3 Rannacher smoothing

The payoff $(K - S)^+$ has a kink at the strike. Rannacher smoothing replaces each of the first two time intervals nearest maturity with two backward-Euler half-steps, giving four half-steps before the Crank–Nicolson updates. This damps initial oscillations from the non-smooth payoff.

3.3.4 Price and delta extraction

For the reported Part A solution, the grid uses $P = 100$ time steps and $M = 300$ spatial steps on $S \in [0, 250]$. The price is interpolated to $S_0 = 140$ from the $t = 0$ grid. Delta is calculated with the central difference

$$\Delta(S_i, 0) \approx \frac{V_{i+1}^0 - V_{i-1}^0}{2\delta S},$$

and interpolated to the same spot value.

4 Results and discussion

4.1 Part A: price and delta comparison

An effective-volatility Black–Scholes calculation provides an approximate closed-form benchmark for the CEV-type diffusion. The effective volatility is defined as:

$$\sigma_{\text{eff}} \approx \left(S_0^{2(\gamma-1)} \frac{1}{T} \int_0^T \sigma(t)^2 dt \right)^{1/2}.$$

This gave $\sigma_{\text{eff}} = 0.1687$ and the estimated Black–Scholes put price 9.1492. Both numerical prices are close to this baseline, and the Black–Scholes value lies inside the Monte Carlo confidence interval.

Table 1. Initial price and delta comparison (4 d.p.).

Quantity	Value
Black–Scholes (estimated) put price	9.1492
Effective volatility σ_{eff}	0.1687
Monte Carlo put price	9.0577
95% confidence interval (price)	[8.7989, 9.3165]
Monte Carlo delta	-0.4009
95% confidence interval (delta)	[-0.4091, -0.3927]
Crank–Nicolson put price	9.1295
Crank–Nicolson delta	-0.4071

Table 1 compares the prices at the specified spot and parameter values. Both the Crank–Nicolson price and the approximate Black–Scholes reference lie within the Monte Carlo price interval, giving a useful comparison of the three calculations at the reported precision.

The Crank–Nicolson delta also lies inside the Monte Carlo 95% interval. The two methods therefore give similar local hedge-ratio estimates at S_0 .

4.2 Additional sensitivities

Using finite-difference bumps (as implemented in the code), we obtained:

$$\Gamma \approx 0.0141, \quad \text{Vega} \approx 49.2514, \quad \Theta \approx -3.8189, \quad \rho \approx -91.0930.$$

Vega and rho are reported per unit change in decimal volatility and interest rate. For example, $\rho \approx -91.09$ implies a first-order price change of approximately -0.91 for a one-percentage-point increase in r . These are local finite-difference estimates, so bump size and simulation variance affect their precision.

4.3 Solution and hedge-profile diagnostics

The path band summarises the dispersion of simulated underlying prices through time, alongside the mean simulated path.

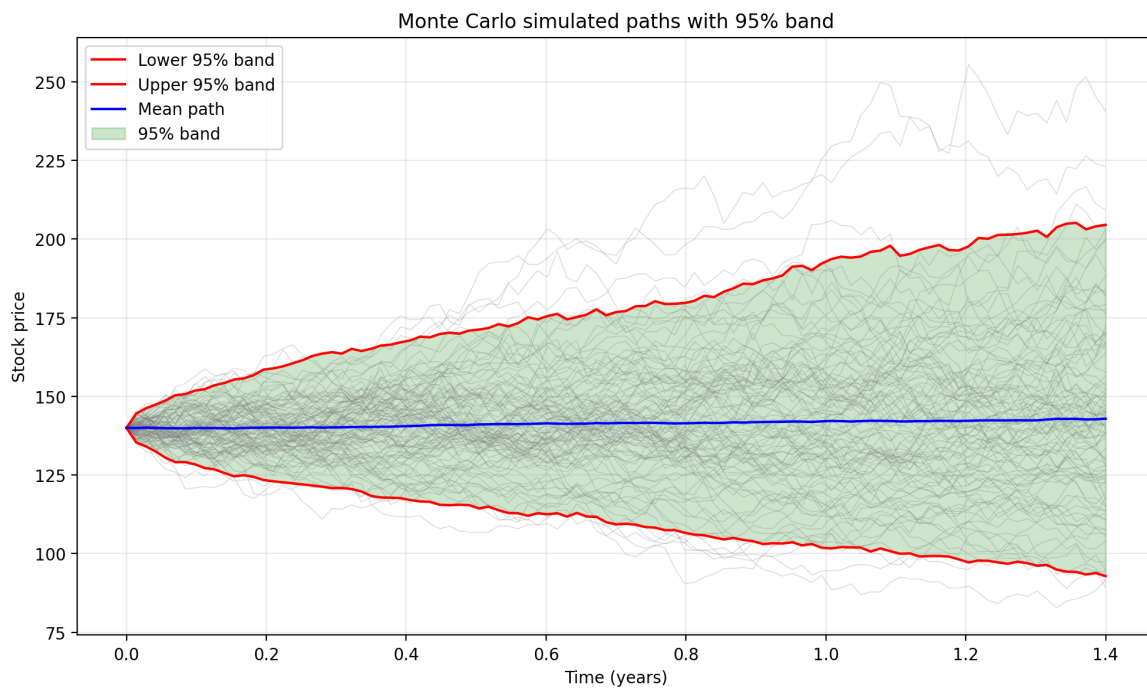


Figure 1. Monte Carlo simulated stock paths with 95% band and mean path.

The value surface shows the numerical solution approaching the terminal payoff. The delta profile moves from approximately -1 at low spot values towards zero at high spot values, following the expected shape of a European put.

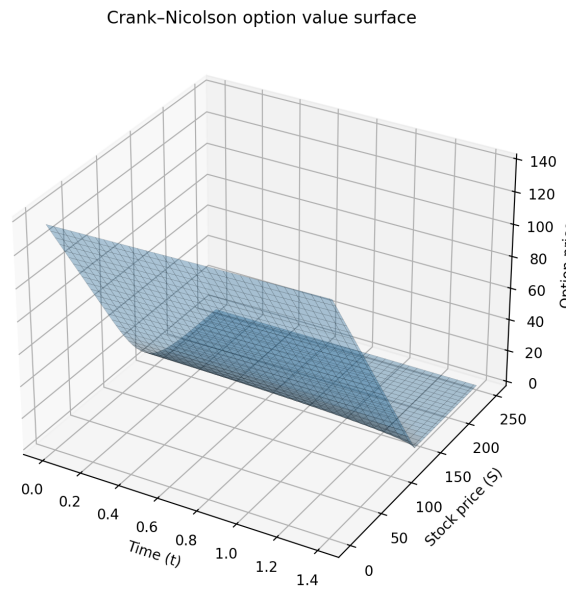


Figure 2. Crank–Nicolson value surface $V(S, t)$ over $S \in [0, 250]$ and $t \in [0, T]$.

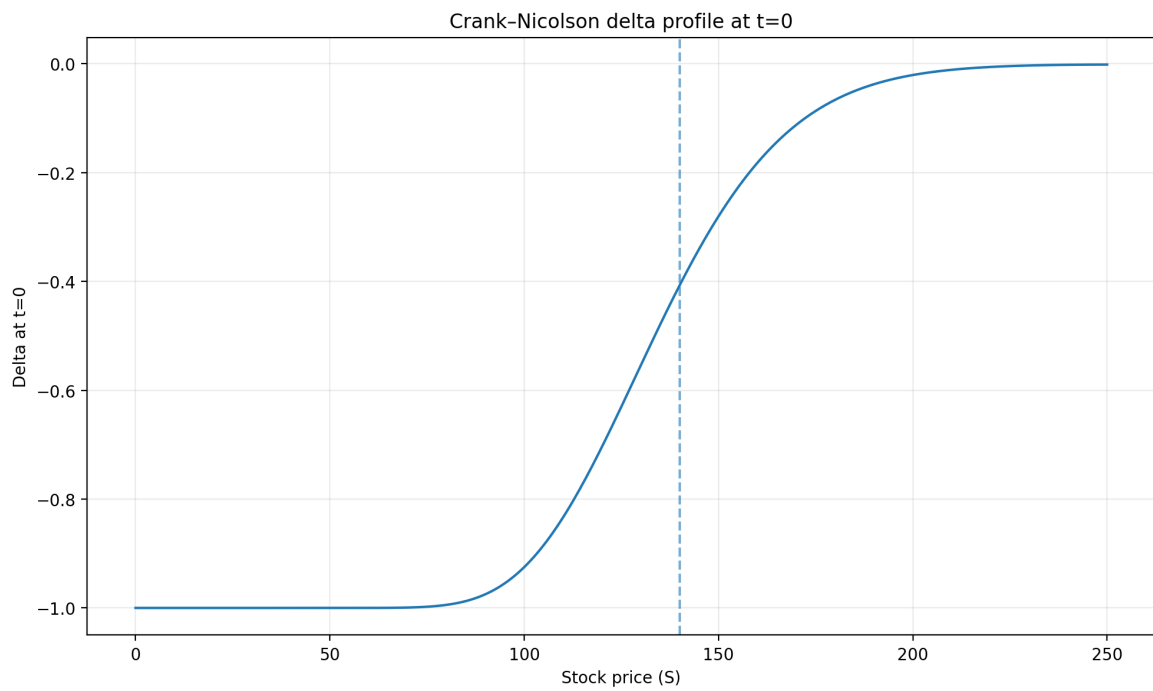


Figure 3. Crank–Nicolson delta profile $\Delta(S, 0)$ across the spatial grid (dashed line at $S = K$).

4.4 Convergence and sampling error

To compare numerical behaviour more directly, we plotted relative errors against high-resolution reference values:

- Monte Carlo errors are computed against a higher-order Monte Carlo estimate using $N_{\text{ref}} = 1,000,000$ paths (generated independently).
- Crank–Nicolson spatial errors fix $P = 500$ and compare against a high-resolution spatial reference with $M_{\text{ref}} = 100,000$.
- Crank–Nicolson temporal errors fix $M = 500$ and compare against a high-resolution temporal reference with $P_{\text{ref}} = 100,000$.

The horizontal axis uses N for Monte Carlo and $M \cdot P$ for Crank–Nicolson. These are refinement proxies, not matched runtimes or operation counts: Monte Carlo path work also depends on the number of time steps.

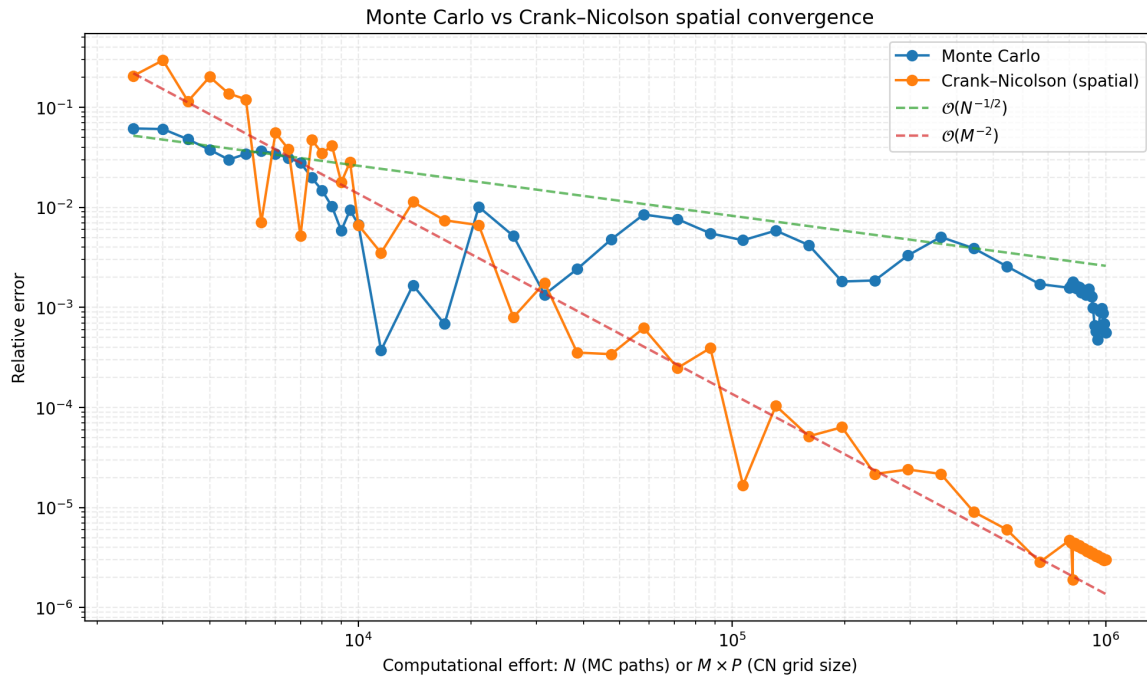


Figure 4. Monte Carlo relative error vs spatial Crank–Nicolson relative error (both measured against high-resolution references).

Figures 4 and 5 are the *primary* numerical comparison in Part A. They show a clear separation between statistical error (Monte Carlo) and discretisation error (Crank–Nicolson). On the right of both plots, at effort around 10^6 , the Monte Carlo relative error sits around the 10^{-3} scale (a few $\times 10^{-4}$ to 10^{-3}), consistent with the slow $\mathcal{O}(N^{-1/2})$ rate.

In contrast, the Crank–Nicolson curves fall much faster. In the spatial refinement plot (Figure 4), Crank–Nicolson reaches around 10^{-5} error already by effort on the order of a few $\times 10^5$, and pushes into the few $\times 10^{-6}$ range by 10^6 , aligning with the $\mathcal{O}(M^{-2})$ guide once the asymptotic regime is reached. The early “spikiness” is expected before the clean second-order regime dominates (and is also influenced by reference-value accuracy and the payoff kink).

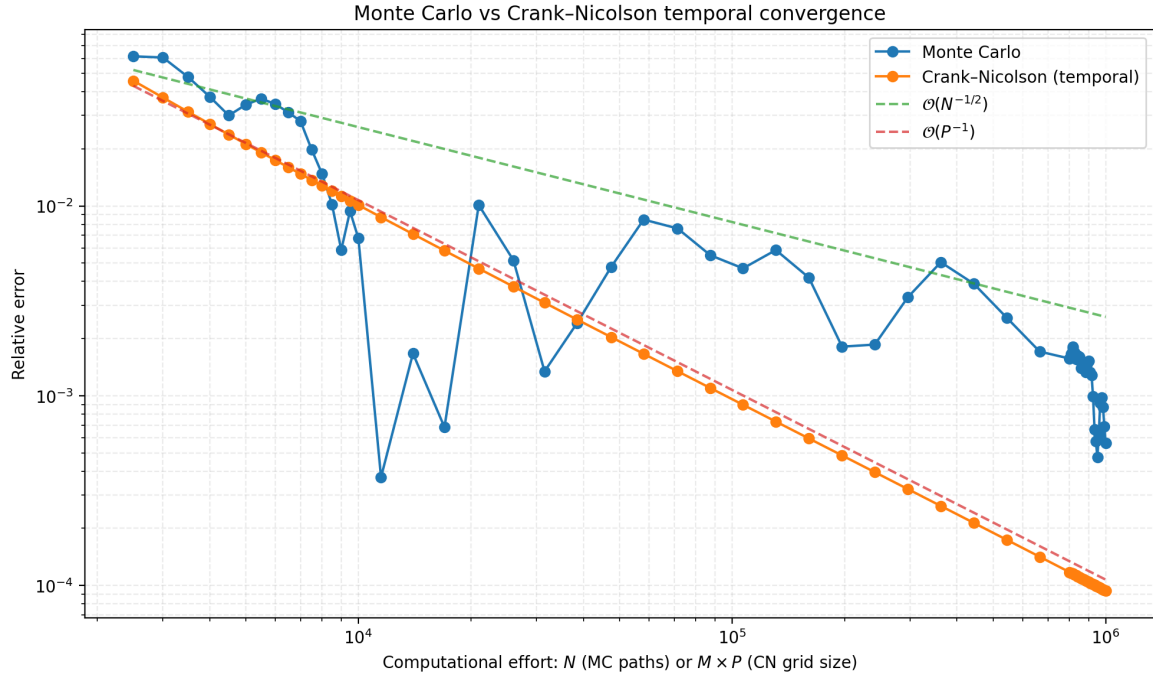


Figure 5. Monte Carlo relative error vs temporal Crank–Nicolson relative error (both measured against high-resolution references).

For temporal refinement (Figure 5), the observed error follows approximately the $\mathcal{O}(P^{-1})$ guide and reaches around 10^{-4} at the plotted effort of 10^6 . Temporal refinement is therefore slower than spatial refinement in this experiment. The comparison shows how grid choice affects the two components of discretisation error.

Figure 6 shows the sampling-error scale, $\hat{\sigma}/\sqrt{N}$. The $N^{-1/2}$ relationship illustrates the reduction in estimated standard error as the number of simulated paths increases.

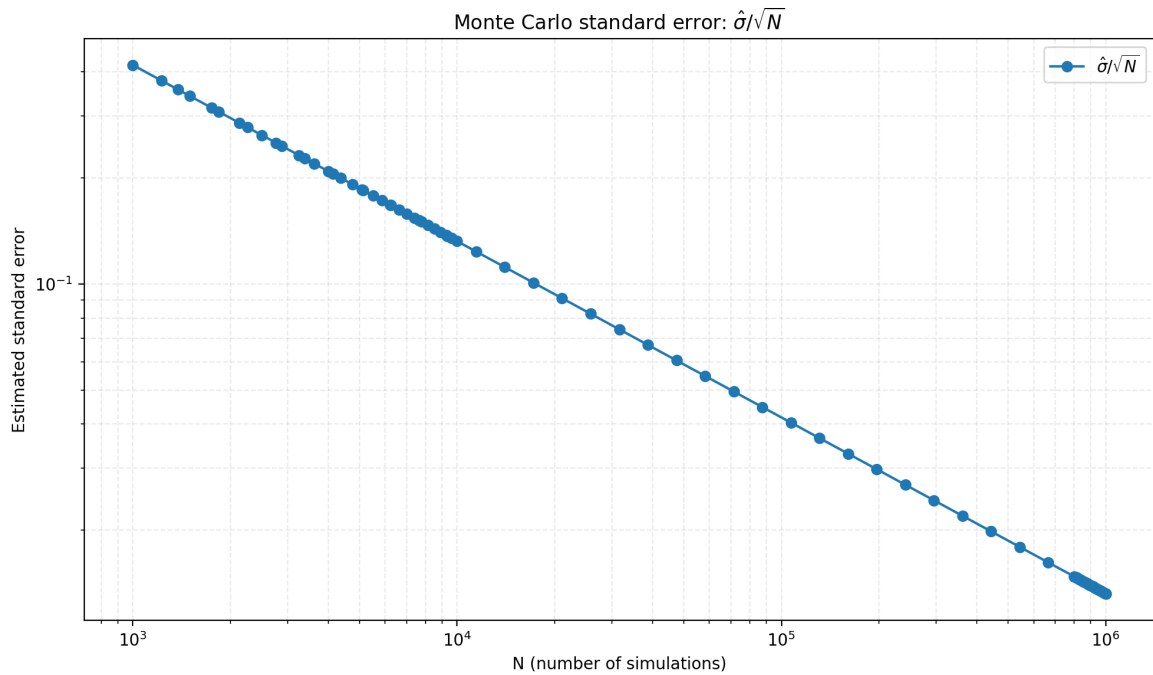


Figure 6. Monte Carlo estimated standard error $\hat{\sigma}/\sqrt{N}$ on log–log axes, illustrating the $N^{-1/2}$ sampling-error scale.

4.5 Part B: option-chain comparison

Part B applies European pricing calculations to the supplied SPY chain under GBM with continuous dividend yield q . For each contract it reports Black–Scholes prices using the supplied implied volatility, MC prices and sampling intervals, CN prices, Black–Scholes Greeks, and model–mid differences.

The run used an SPY spot level of 682.85 and a dividend yield $q = 1.050\%$. The cleaned chain contained 947 contracts across 10 expiries and strikes from 510 to 750.

Table 2. SPY chain summary (Part B).

Measure	Value
Options priced	947
Calls / Puts	280 / 667
Strike range	510 – 750
Implied vol range (mean)	10.7% – 53.5% (22.48%)
Dividend yield q	1.050%
Market mid mean (std)	3.2851 (3.1712)
BS mean (std)	3.0320 (3.0822)
MC mean (std)	2.9835 (3.0542)
CN mean (std)	3.0282 (3.0689)

The BS and CN aggregate prices in Table 2 are close under their common model assumptions. Monte Carlo has a slightly lower mean price. The distribution plots and contract-level sampling intervals provide further detail on the differences between the methods.

Table 3. Model–mid difference metrics vs market mid (Part B).

Model	Mean (model-mid)	MAE	MaxAbs
BS	-0.2531	0.3065	1.0717
MC	-0.3016	0.3378	1.1653
CN	-0.2569	0.3102	1.1130

Table 3 summarises model prices minus quoted market mids. The mean difference describes the overall direction of the gap, MAE measures its typical magnitude, and MaxAbs identifies the largest absolute difference. These statistics complement the aggregate price summary.

4.6 Implied volatility and risk structure

For the SPY chain, the implied volatility skew and term structure provide immediate context for risk. Figure 7 shows the skew across expiries, while Figure 8 summarises how Delta/Gamma/Vega exposures vary by strike and maturity.

Figure 7 shows a pronounced *downside skew*: implied volatilities are substantially higher at lower strikes (deep OTM puts) and decline toward the ATM region. Across the higher-strike region the curves flatten and even show a slight right-tail uptick for some maturities, consistent with a

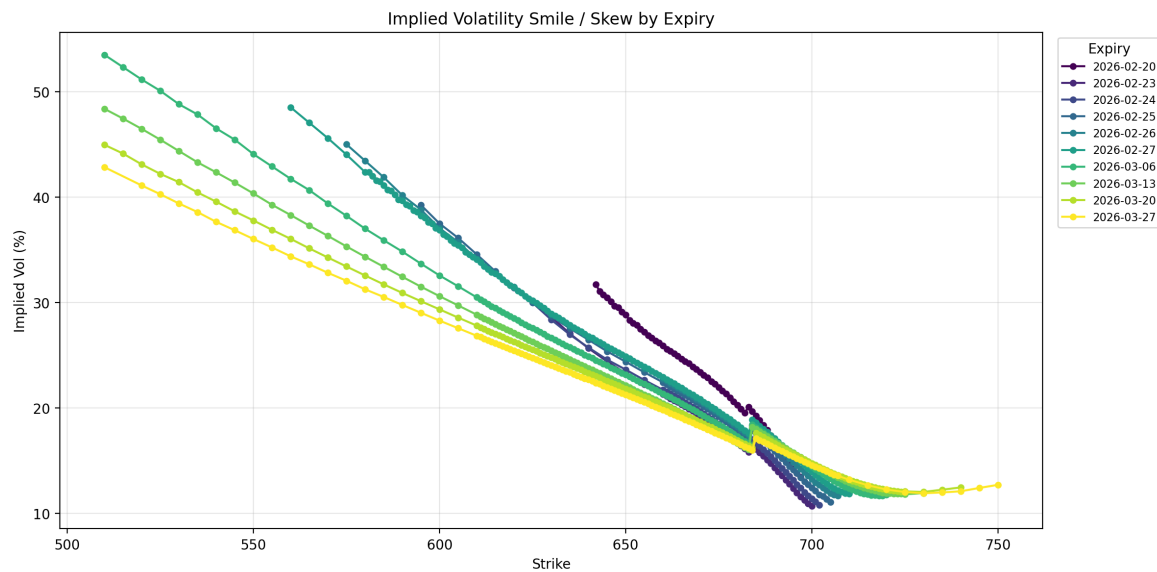


Figure 7. SPY implied volatility smile/skew by expiry (Part B).

skew/smile shape rather than a perfectly flat volatility surface. The separation between expiry curves reflects term structure: overall level and slope vary by maturity, which is precisely why a chain-level engine needs to handle expiry-by-expiry inputs rather than a single volatility number.

Figure 8 matches standard option-risk intuition and also acts as a sign/convention check on the engine outputs. Gamma peaks around the ATM region (near spot), with the highest peaks occurring in the shorter-dated expiries, while longer-dated expiries exhibit lower gamma but higher vega. The vega curves rise with maturity and peak around ATM, reflecting greater sensitivity of longer-dated options to volatility changes.

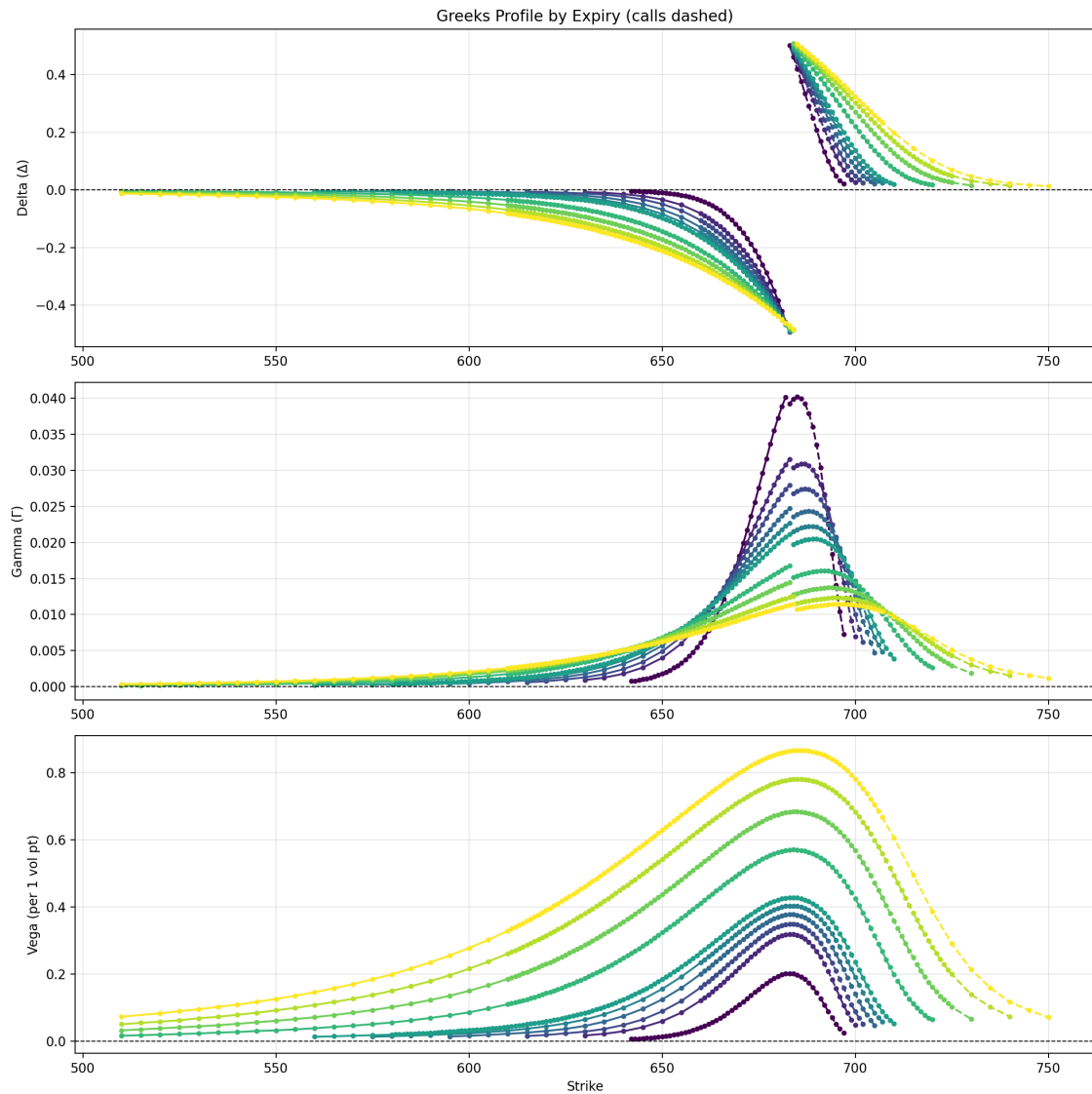


Figure 8. SPY Greeks profiles by expiry (Black–Scholes Greeks; calls dashed) (Part B).

4.7 Model–market diagnostics

Figure 9 displays the distribution of model–mid differences for the three methods. The distributions are centred below zero, and the BS and CN profiles are similar. Their spread shows how the magnitude of the difference varies across contracts.

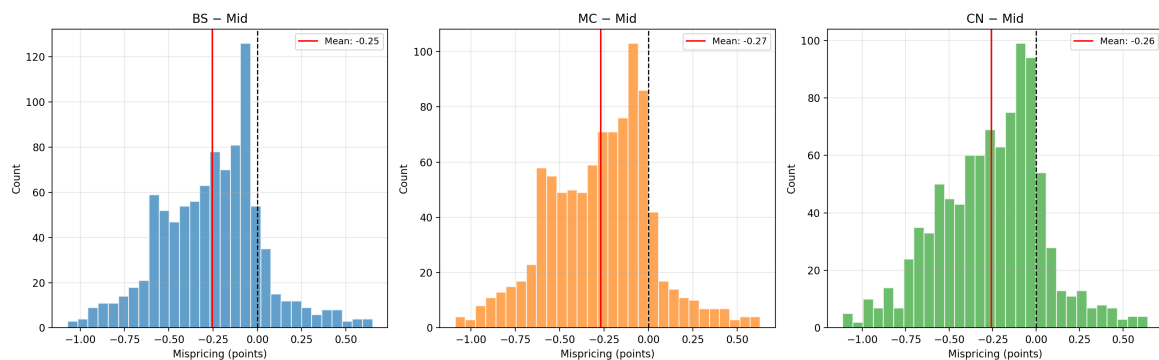


Figure 9. Model–mid difference distributions for BS, MC and CN (Part B).

Figure 10 places market mids alongside Monte Carlo estimates and their sampling intervals for a near-the-money subset. Several market mids lie above the intervals. The plot allows the model–mid gaps and the scale of Monte Carlo sampling variation to be inspected together.

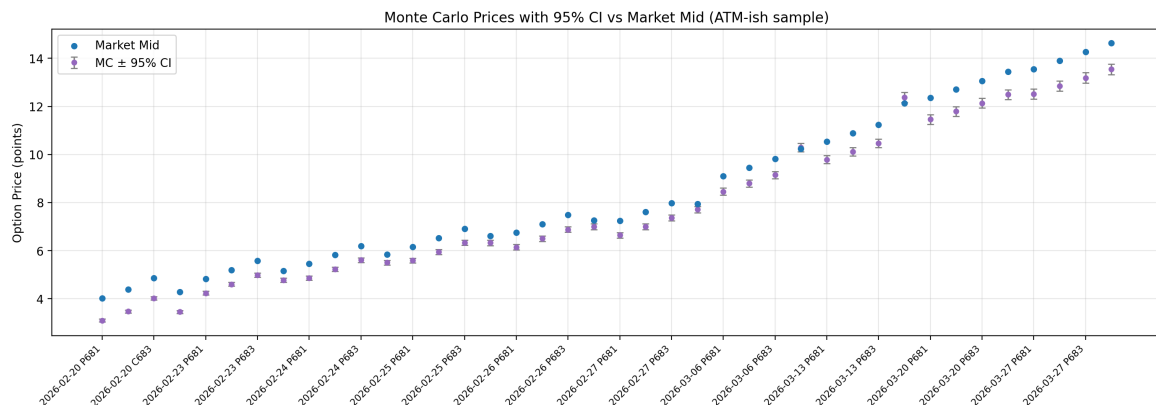


Figure 10. Monte Carlo price estimates and 95% sampling intervals alongside market mids for a near-the-money subset (Part B).

4.8 Scenario price changes

The scenario analysis selects six contracts nearest to the money and reprices them under spot shocks of $\{-10\%, -5\%, 0\%, +5\%, +10\%\}$ and volatility shocks of $\{-5\text{pp}, -2\text{pp}, 0, +2\text{pp}, +5\text{pp}\}$. Here pp denotes absolute percentage points of implied volatility. Each repriced value is compared with its stored Black–Scholes reference price. The chart sums these differences using one unit of each option and no contract multiplier.

Figure 11 arranges the scenario outputs by spot shock and volatility shock. The colour scale and cell values show the summed price difference relative to the stored references for each combination.

This view extends the local Greek profiles to simultaneous finite changes in spot and volatility. It provides a compact illustration of the scenario-reporting component of the pricing workflow.

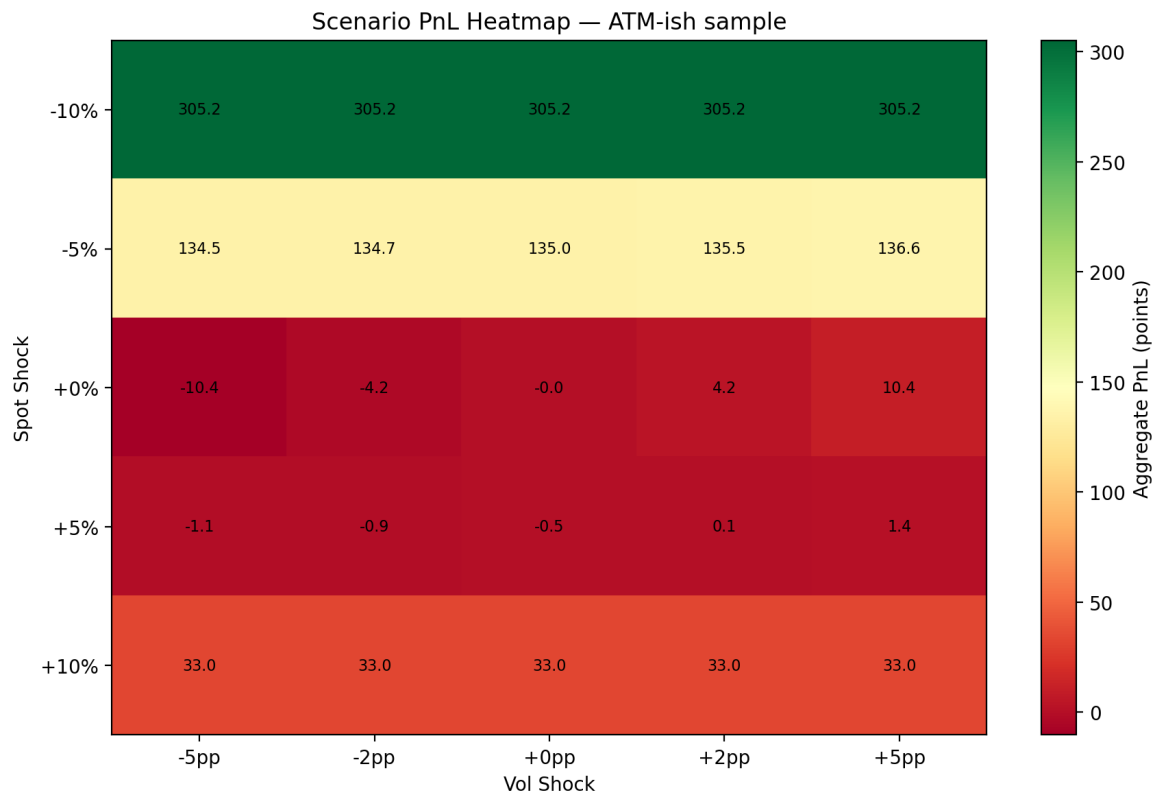


Figure 11. Illustrative scenario price differences for a near-the-money subset under joint spot and volatility shocks, summed in quote units.

5 Limitations and further work

- **Model setting.** Part A uses a CEV-type diffusion with an effective-volatility Black–Scholes comparison. Part B assumes European exercise, GBM dynamics and continuous dividends. These choices define the scope of the pricing exercise.
- **Numerical resolution.** Monte Carlo results depend on path count and time step, while finite-difference results depend on grid spacing, domain boundaries and payoff treatment. Sampling intervals describe the simulation component of uncertainty.
- **Market snapshot.** The chain represents one underlying at one observation time. Quote timing, bid–ask spreads and contract conventions provide context when comparing model prices with market mids.
- **Scenario scope.** The heatmap uses a small near-the-money subset and one unit per option. Position sizes, contract multipliers, transaction costs and dynamic hedging would be additional inputs to a trading-book analysis.

Further work could compare convergence across more parameter settings, benchmark runtime directly and extend the market analysis to additional observation dates. Alternative exercise and dividend treatments would broaden the range of contracts supported by the pricing engine.

6 Conclusion

Monte Carlo and Crank–Nicolson produce similar Part A prices and deltas at the reported precision. The comparison illustrates the distinction between sampling error and discretisation error, while the value surface, delta profile and refinement experiments show how the numerical methods behave.

The option-chain application extends these methods into a pricing and risk-reporting workflow covering 947 contracts. Price comparisons, volatility profiles, Greeks and scenario outputs bring the numerical results together in a practical analytical format. The project demonstrates the progression from a mathematical pricing model to implementation, sensitivity analysis and clear reporting.

References

- [1] John Armstrong. Lecture notes for 7CCMFM06, Computational and Numerical Methods in Mathematical Finance. <https://keats.kcl.ac.uk/course/view.php?id=119832>, 2025.